



Complete Css Notes



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Notes



Selectors

Concepts

Selectors are used to **target HTML elements** so that styles can be applied.

👉 Types:

- Universal (`*`) → targets all elements
- Element (`p`) → targets specific tags
- Class (`.class`) → reusable styling
- ID (`#id`) → unique element
- Group (`h1, p`) → multiple elements

👉 Example:

```
* {  
  margin:0;  
  padding:0;  
}  
  
.card {  
  background:blue;  
}  
  
#header {  
  font-size:20px;  
}
```

Colors & Units

Concepts

Used to define **colors and sizing** in CSS.

👉 Color Types:

- Named → `red`
- HEX → `#ff5733`
- RGB → `rgb(255,0,0)`
- RGBA → transparency support

👉 Units:

- `px` → fixed
- `%` → relative
- `em` → parent-based
- `rem` → root-based

👉 Example:

```
color:red;
background-color:#ff0000;
width:50%;
```

Text Styling

Used to control **text appearance and spacing**.

👉 Example:

```
text-align:center;
text-transform:uppercase;
text-decoration:underline;
line-height:1.5;
letter-spacing:2px;
word-spacing:5px;
```

Fonts

Used to style **text font and typography**.

👉 Example:

```
font-family:Arial,sans-serif;
font-size:16px;
font-weight:bold;
font-style:italic;
```

👉 Google Fonts:

```
<link href="https://fonts.googleapis.com/css2?family=Poppins&display=swap" rel="stylesheet">
```

```
body {  
  font-family: 'Poppins', sans-serif;  
}
```

Box Model

Concepts

Every element follows:

Content → **Padding** → **Border** → **Margin**

👉 Example:

```
width:200px;  
height:100px;  
padding:10px;  
border:2px solid black;  
margin:20px;  
box-sizing:border-box;
```

Background

Used to style **element backgrounds**.

👉 Example:

```
background-color:red;  
background-image:url('image.jpg');  
background-repeat:no-repeat;  
background-position:center;  
background-size:cover;
```

Display Property

Controls how elements are **rendered on screen**.

👉 Types:

- `block` → full width
- `inline` → no width/height
- `inline-block` → hybrid
- `none` → hidden

👉 Example:

```
display:block;
display:inline-block;
```

⚡ Flexbox

🧠 Concepts

Used for **1D layouts (row OR column)**.

👉 Example:

```
.container {
  display:flex;
  flex-direction:row;
  justify-content:center;
  align-items:center;
  flex-wrap:wrap;
  gap:10px;
}
```

🧱 CSS Grid

🧠 Concepts

Used for **2D layouts (rows + columns)**.

👉 Example:

```
.container {
  display:grid;
  grid-template-columns:repeat(3,1fr);
  grid-template-rows:auto;
  gap:10px;
```

```
place-items:center;
}
```

Positioning

Controls **element placement**.

👉 Types:

- `static` → default
- `relative` → from itself
- `absolute` → from parent
- `fixed` → screen fixed
- `sticky` → sticks on scroll

👉 Example:

```
position:absolute;
z-index:10;
```

Pseudo Classes

Used for **element states**.

👉 Example:

```
button:hover {
  background:red;
}

input:focus {
  border:2px solid blue;
}

li:nth-child(2) {
  color:green;
}
```

Pseudo Elements

Used to style **specific parts of elements**.

👉 Example:

```
p::before {
  content:"🔥";
}

p::after {
  content:"✓";
}

::selection {
  background:yellow;
}
```

Transitions

Used for **smooth changes**.

👉 Example:

```
button {
  transition:all 0.3s ease-in-out;
}

button:hover {
  background:red;
}
```

Transform

Used to **move or change shape**.

👉 Example:

```
transform:translate(50px,0);
transform:rotate(45deg);
transform:scale(1.2);
transform:skew(20deg);
```

Animations

Used to create **motion effects**.

👉 Example:

```
@keyframes slide {  
  from { transform:translateX(0); }  
  to { transform:translateX(100px); }  
}  
  
.box {  
  animation-name:slide;  
  animation-duration:2s;  
  animation-iteration-count:infinite;  
}
```

Responsive Design

Concepts

Makes websites **adapt to different screen sizes**.

👉 Media Query:

```
@media (max-width:768px) {  
  body {  
    background:red;  
  }  
}
```

👉 Breakpoints:

- 576px → mobile
- 768px → tablet
- 992px → laptop
- 1200px → desktop

👉 Width Control:


```
max-width:1200px;  
min-width:300px;
```